

Dhruv Tomar

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Education

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science

Bachelor of Science in Integrative Biology

Expected Graduation: May 2027

GPA: 3.93/4.00

Relevant Coursework: Algorithms & Models of Computation, Linear Algebra, Applied Machine Learning, Probability & Statistics, Cells, Tissues & Development, Ecology, Physiology, Genetics, Elementary Organic Chemistry I

Honors and Awards

Illinois for Illinois (I4I) Study Abroad Scholarship 2025

James Scholar Honors Program 2025

Dean's List (x4) 2023, 2024, 2025

Research Experience

Undergraduate Research Assistant

August 2024 - Present

University of Illinois Urbana-Champaign, Department of Natural Resources & Environmental Sciences

- Investigating the effects of light pollution on bird flight behavior using 3D object tracking in thermal videos in the Van Doren Lab of Migration Biology.
- Optimized background subtraction methods for thermal video object tracking.
- Trained and optimized a YOLOv8 machine learning model to improve object detection performance.
- Performing extrinsic calibration of thermal cameras and analyzing over 10,000 tracks using MATLAB and Python (NumPy, pandas, SciPy, TensorFlow).

Research Intern

June 2025 - August 2025

University of California San Diego, Department of Computer Science

- Summer 2025 REU Intern in the Engineers for Exploration program.
 - Collaborated with National Geographic explorers on two large-scale bioacoustic data mining projects, characterizing soundscapes with little to no labeled data.
 - Applied template matching and unsupervised deep learning to build a Graph Neural Network pipeline uncovering latent links between audio features and environmental variables.
 - Pioneered a modular Python package enabling scalable, cross-dataset bioacoustic analysis and integration into future ecological research workflows using libraries such as PyTorch, TensorFlow, and Librosa.
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Teaching & Professional Experience

Lead Engineering Learning Assistant

January 2026 - Present

Engineering Learning Assistant

August 2025 - December 2025

University of Illinois Urbana-Champaign, Grainger College of Engineering

- Teaching and mentoring a section of ~20 Computer Science students in a first-year engineering orientation (ENG 100) on academic, career, and campus life topics.
- Facilitating student learning and community-building using curated lesson materials.
- Attending weekly instructional meetings to review pedagogy and curriculum design.

Intern, Cretorial Media Services

June 2024 - August 2024

- Collaborated with a startup team to develop a comprehensive dictionary.
- Utilized generative AI APIs to produce accurate and engaging encyclopedia entries focused on animal species and their ecological role.
- Ensured content aligned with educational and conservation standards for target audiences, including students and naturalists.

Conference Presentations

Undergraduate Research Symposium, Urbana, IL, April 2025. Tomar, D. "3D Tracking in Thermal Videos to Analyze the Effects of Light Pollution on Birds" (poster).

Publications and Presentations

Perry, S., Zhang, T., Kamboj, S., Jajodia, A., Tomar, D., & Kastner, R. (2025). Studying Domain Shifts in Bioacoustic Recording Style With Information Theory. Available at SSRN (Preprint No. 5510368).

Tomar, D. (2025). Factors affecting migratory bird-window collisions and effective mitigation strategies. *Double Helix Digest*, 1(1), 9-16.

Extracurricular Experiences

Vice President, Treasurer, Illini Wildlife and Conservation Club

May 2024 - Present

- Organize monthly general meetings and invite professors to discuss their research.
- Manage funds and budget for a 50+ member club.
- Participate in philanthropy such as turning off lights around campus to conserve energy and picking up litter around campus.

Bird Collision Monitor, UIUC Bird Strike Survey

September 2023 - Present

- Survey 20+ buildings on campus during the bird migration season several times a week for bird-window collisions following survey protocols and log them via ArcGIS QuickCapture.
- Collect specimens for use in teaching and research.

Professional Memberships

American Ornithological Society (2023-Present)

Association for Computing Machinery (2025-Present)

The Wildlife Society (2023-Present)

Technical Skills

Programming languages: C++, Python, R, MATLAB, Java, JavaScript, Verilog

Tools & Systems: Git, VS Code, Docker, PostgreSQL, Bootstrap 5, MongoDB

Libraries: NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Matplotlib, Librosa